

Replicating ReKG-MCTS

Reinforcing LLM Reasoning on Knowledge Graphs via Training-Free Monte Carlo Tree Search

Paper

ACL 2025 (Song et al.)

Model

DeepSeek-V3

Dataset

RoG-WebQSP

Why Knowledge Graphs + LLMs?

The Problem with LLMs

LLMs suffer from **hallucination and knowledge inconsistency**

What KGs Provide

Knowledge Graphs (KGs) provide **structured, reliable factual information**

The Core Challenge

Multi-hop reasoning over KGs requires **precise pathfinding**

Limitation of Existing Methods

Existing methods use **static heuristics** (beam search, greedy) that **cannot adapt** based on path quality

The Solution

Monte Carlo Tree Search (MCTS) **dynamically balances exploration vs exploitation**

How ReKG-MCTS Works

Four phases of MCTS:

1

Node Selection

UCB1 criterion:

$$UCB(C) = \frac{v}{n} + c\sqrt{\frac{\ln(n_{parent})}{n}}$$

2

Path Expansion

LLM selects **top-k relevant KG relations** to explore

3

Simulation (MC Rollout)

LLM scores **path relevance** (0.0-1.0)

4

Backpropagation

Update **visit counts and rewards** up the tree

LLM serves dual roles:

Policy Model

Selects which relations to explore

Value Model

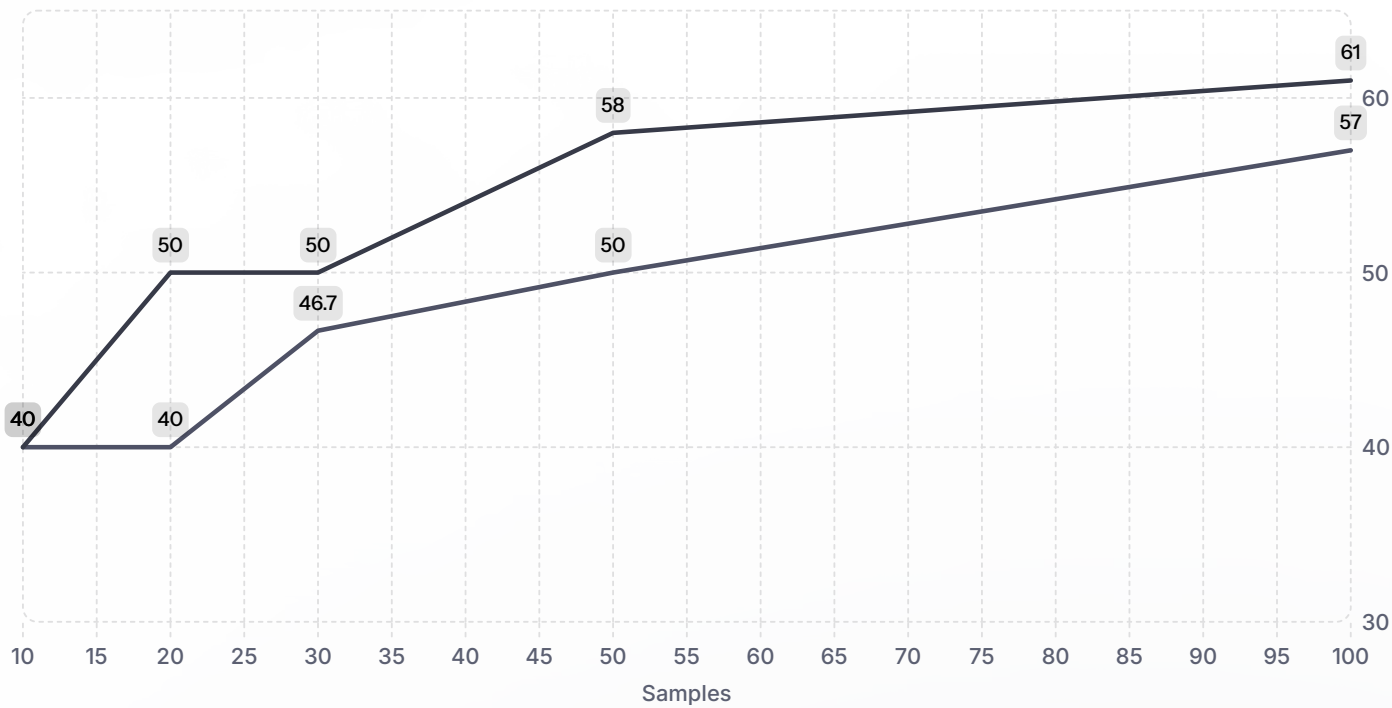
Scores how useful a path is for answering the question

Experimental Results (Hits@1)

Method	10 samples	20 samples	30 samples	50 samples	100 samples
CoT Baseline	40.00%	50.00%	50.00%	58.00%	61.00%
ReKG-MCTS	40.00%	40.00%	46.67%	50.00%	57.00%

📌 **Key observation:** The gap between MCTS and CoT **narrows as sample size increases** (0% gap at 10 → 4% gap at 100), showing MCTS gains advantage on harder multi-hop questions.

— ReKG-MCTS — CoT Baseline



Findings & Takeaways

Why MCTS trails CoT in our setting

- We use **pre-extracted KG subgraphs** (~5K triples/question) instead of full Freebase (1.9B triples)
- Simplified path evaluation prompt introduces **noisy reward signals**
- Paper uses **carefully tuned few-shot prompts** (Appendix E)

What we learned

- Successfully implemented **all 4 MCTS phases** from scratch
- MCTS shows **increasing advantage** on complex multi-hop questions
- Full performance requires **complete KG access** via SPARQL endpoint



Future work: Integrate live Freebase SPARQL endpoint + few-shot path evaluation prompts